Sudarshan A

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Full-Stack Engineer (Node.js · Python · React · TypeScript) with hands-on experience shipping production-grade real-time systems and AI-powered applications. Built a collaborative coding platform with a custom WebSocket relay and dual-auth system, and an AI agent that turns natural conversation into structured search over 3,397 government records using a hybrid RAG pipeline. Strong in REST API design, real-time architecture, and LLM/RAG integration. Seeking a full-time SWE role to build scalable, user-facing products.

TECHNICAL SKILLS

Frontend: React.js, Next.js, TypeScript, JavaScript (ES6+), HTML5, CSS3, TailwindCSS, Context API, Hooks
Backend: Node.js, Express.js, REST APIs, WebSockets, JWT, NextAuth.js, bcrypt, Zod, Rate Limiting
Database: PostgreSQL, MongoDB, MySQL, SQLite, Redis (Upstash)- schema design, aggregation
Real-time: Y.js CRDT, WebRTC, Socket.io, IndexedDB persistence
AI/ML: LLM Prompt Engineering, RAG Pipelines, LangChain, Vector Databases (ChromaDB), Hybrid Search (BM25 + Semantic), Sentence Transformers, Agentic Workflows, Gemini, Groq
Tools: Git, GitHub, Postman, Vercel, AWS (basics), Linux

PROJECTS

S-Code — Real-Time Collaborative Coding Platform [Live] | Next.js . TypeScript . MongoDB . Redis . Y.js . WebSockets

- Built a custom WebSocket relay server (Express + ws) that forwards binary Y.js CRDT messages across all room connections - achieving <200ms sync for 7+ concurrent users
- Engineered a context-aware AI chat sidebar that polls the Monaco editor every second via EditorContext, appends the live code (capped at 10,000 chars) to each Gemini 2.5 Flash Lite prompt - giving users code-specific answers
- Designed a dual auth system: custom JWT flow (HS256, 15-min access + 7-day rotating refresh tokens, HTTP-only cookies) unified with NextAuth.js OAuth (Google + GitHub) via a single withAuth middleware
- Implemented a room state machine (scheduled - live - ended - saved) with enforced transition validation at the API layer - invalid transitions are rejected before any DB write
- Secured the platform with 11 tiered Upstash Redis rate limiters across 5 risk levels (auth endpoints keyed by IP, room/AI/code endpoints by userId),dual-tier AI limits(10/min+50/hr),CSRF origin validation on all mutating routes

YojnaSaathi - AI-Powered Government Scheme Discovery Platform | Next.js · FastAPI · LangChain · ChromaDB · Clerk

- Architected a RAG pipeline over 3,397 Indian government schemes, combining semantic search with BM25 via Reciprocal Rank Fusion, which improved exact-match retrieval for scheme names/abbreviations
- Built a multi-turn conversational agent that extracts structured user profiles from natural language via LLM, using a sufficiency scorer to decide when to search vs. ask follow-ups - eliminating unnecessary form-filling
- Switched embedding provider from Google to Sentence Transformers after hitting API rate limits, and built an llm_factory.py abstraction so the LLM provider (Groq, swapped in after Gemini deprecation) can be changed via .env with no code changes

FlashMind - AI-Powered Spaced Repetition Study App | Next.js. TypeScript. PostgreSQL. Gemini 2.5 Flash. Groq

- Built a full-stack AI study tool that auto-generates 2-4 atomic flashcards per input using Gemini 2.5 Flash, with automatic Groq fallback on 503 errors - ensuring zero-downtime AI generation
- Implemented the FSRS spaced repetition algorithm (ts-fsrs) with a PostgreSQL revlog table for per-user scheduling based on stability, difficulty, and retrievability-reducing required review sessions by ~40% vs fixed-interval systems

EXPERIENCE

Java Full-Stack Intern - Tap Academy, Bengaluru Jan 2026 - Present

- Developing Core Java proficiency covering OOP principles, exception handling, and Collections Framework through a structured full-stack curriculum
- Advancing into Jakarta EE for enterprise web development - Servlets, JSP and JPA

EDUCATION

Bachelor of Engineering (B.E). in Computer Science and Engineering **CGPA: 8.41/10**
Shree Devi Institute of Technology | *Expected Graduation: Jun* 2026

CERTIFICATIONS & KEY ACHIEVEMENTS

- OCI AI Foundations — Oracle
- Full-Stack Web Development — Angela Yu / Udemy
- AWS Cloud Technical Essentials — Amazon / Coursera